25900743 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCBCH100 - BIOCHEMISTRY-THE SCIENCE OF LIFE

(2024 ADMISION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Which of the following is NOT a feature of eukaryotic cells? [U] / [CO1]

   a). a) Membrane-bound organelles b). b) Nucleoid region

   c). c) Cytoskeleton d). d) Nuclear membrane

2. Identify the monosaccharide among these: [U] / [CO2]

   a). Sucrose b). Glycogen

   c). Glucose d). Maltose

3. Atom most commonly involved in hydrogen bonding [U] / [CO1]

   a). Oxygen b). Sodium

   c). Calcium d). All of the above

4. Which of the following hetro polysaccharides is known for its role in the skin and joint lubrication? [K] / [CO2]

   a). A chondroition sulfate b). Hyaluronic acid

c). Hetarin

d). Glycogen

5. What is the function of ribosomal RNA (rRNA)? [K] / [CO4]

a). Catalyzes the attachment of amino acids to tRNA

b). Transfers amino acids to the ribosome

c). Forms the structural and catalytic components of ribosomes

d). Carries genetic information from DNA

[5x1=5]

Part B

Fill in the Blanks

Answer **All** Questions

Each Question carries **2** marks

6. The Kw value for water at 25°C is _____, but at higher temperatures, this value _____. [U] / [CO1]

7. As the concentration of hydroxide ions increases, the pOH value ___. [U] / [CO1]

8. The formation of ring structure in monosaccharides creates a new center of asymmetry,this results in 2 isomeric forms called_____ [U] / [CO2]

9. Which is the storage form of glucose in animals. [K] / [CO2]

10. The structure of nucleotides allows them to _____ information in genetic material. [An] / [CO4]

[5x2=10]

Part C

Short Answer Questions

Answer any **5** Questions

Each Question carries **3** marks

11. Explain Mono unsaturated fatty acid and give one example with structure. [A] / [CO4]

12. Write a note on peptide bond. [K] / [CO3]

13. Justify that structural features allow DNA to carry hereditary information. [E] / [CO4]

14. In what ways does biochemistry influence nutrition? [U] / [CO1]

15. 3. Differentiate between monosaccharides, disaccharides, and polysaccharides, with examples of each. [U] / [CO2]

16. What is Biochemistry? [U] / [CO1]

17. Summarize different classes of lipids. [U] / [CO4]

[5x3=15]

Part D

Short Essay Type Questions

Answer any **4** Questions

Each Question carries **5** marks

18.	Explain the significance of the pKa value in the Henderson-Hasselbalch equation. How does the pKa value affect the pH of a buffer solution?	[U] / [CO1]
19.	Which are the main types of RNA and their functions?	[K] / [CO1]
20.	What are amino acids? Mention the one letter and three letter codes of twenty standard amino acids.	[K] / [CO3]
21.	Explain the concept of zwitterion and its main properties.	[U] / [CO3]
22.	Differentiate the structure of triacylglycerol. How does its composition relate to its role in energy storage?	[An] / [CO4]
23.	Explain the structure and complementary base pairing of DNA.	[A] / [CO4]

[4x5=20]